Huey Phan

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Summary

AI/ML Engineer with 5+ years of software engineering experience and an M.S. in Artificial Intelligence (UT Austin, expected 2027). Builds and ships production ML systems, including a LightGBM forecasting pipeline that lifted at-risk detection from 0.80 to 0.95 ROC-AUC, alongside deep learning research in CNNs, transformers, and generative models. Strong full-stack foundation (Python, Go, React, Oracle) for taking models from notebook to production API.

Education

The University of Texas at Austin, MS in Artificial Intelligence – Austin, TX — Jan 2025 – May 2027

California State University, San Bernardino, BS in Computer Science – San Bernardino, CA — Sept 2014 – Dec 2017

Experience

Programmer Analyst, Pasadena City College – Pasadena, CA — Sept 2021 – present

- Built a machine learning pipeline (Python, LightGBM, scikit-learn) to forecast course enrollment and cancellation risk from 10 years / ∼104K historical Oracle records, cutting section-level prediction error by 17% and improving at-risk detection accuracy from 0.80 to 0.95 ROC-AUC versus a naive historical baseline.
- Engineered a leak-free feature pipeline with time-based (not random) train/test validation to prevent data leakage in temporal forecasting, a common failure mode that silently inflates model accuracy in production ML systems.
- Designed the model-serving architecture (FastAPI microservice behind the existing Go backend) to expose enrollment predictions to the scheduling department's React application via REST.
- Designed and implemented a microservices architecture using Go (Golang) and React to handle 40k+ concurrent users for student data portals.
- Integrated GraphQL APIs with Oracle Database to reduce query time by 60% and improve real-time data accessibility.
- Spearheaded the migration of legacy monolithic systems to containerized Docker and Kubernetes (K8s) clusters, enhancing system scalability and deployment speed.

Software Developer, Cloudsquare – Woodland Hills, CA — Jan 2020 – Sept 2021

- Developed custom Salesforce applications using Apex, Lightning Components, and Visualforce, improving business process efficiency and user workflows.
- Built a real-time face detection pipeline using TensorFlow Lite and Raspberry Pi to demonstrate edge-AI capabilities for client demos.
- Integrated complex third-party APIs (Microsoft Graph, Plaid, Reuters) into scalable backend systems.

Database Administrator, ADR Services, Inc. – Los Angeles, CA — July 2018 – Jan 2020

- Optimized query execution plans for Oracle databases, reducing table scans and boosting retrieval speeds by 60%.
- Developed Python-based ETL (Extract, Transform, Load) pipelines to automate data governance and integrity checks.

Projects

SuperTuxKart AI — Jan 2025

Built CNN/Transformer models for classification, segmentation, detection; trained an autonomous driving agent.

Autoregressive Image Generation June 2025

Decoder-only Transformer that generates images token-by-token from a quantized codebook (BSQ/VQ). Trained end-to-end on SuperTuxKart frames.

Class Search Portal & Hub Aug 2024

React + Go middleware over vendor GraphQL/REST. Card-based results, filters that scroll with the cursor. Serving 40k+ users.

Publications

Comparing CNN, Recurrent, Hybrid, and Transformer Models for Multi-Label ECG Classification on PTB-XL Aug 2026
Huey Phan
10.31224/7898 (engrxiv)

The Challenge of Debiasing NLI Models: Why Hypothesis-Only Confidence is Insufficient Jan 2026
Huey Phan
10.31224/6210 (engrxiv)

Predicting ZIP Level Housing Values in Los Angeles County: A Machine Learning Approach Using Public Data May 2026
Huey Phan
10.20944/preprints202605.2102.v1 (preprints)

Skills

Languages: Python, Javascript, Golang, Java, C#, SQL, Apex

ML & Deep Learning: PyTorch, scikit-learn, LightGBM, HuggingFace, LoRA, CNNs, Transformers, CLIP, Quantization

Data & MLOps: pandas, NumPy, FastAPI, Oracle, feature engineering, time-based validation

Infrastructure: Docker, Kubernetes, AWS, Azure